

ScrapStitch

Group Number: Cohort 1, Group 4

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I. Introduction

In today's digital era, the way we preserve our treasured memories has evolved from classic methods like scrapbooking and journaling to more advanced digital formats. Smartphones enable us to effortlessly capture and revisit our memories through photos, videos, and social media posts, offering a modern twist to memory keeping. However, the efficiency of digital mediums comes with some unique downsides. The increase in information caused by digitalisation may cause digital clutter, hindering the user's ability to meaningfully store and recall past memories, as doing so would require the user to organise, analyse and interpret the vast quantity photos, videos, posts, music and data being recorded at any point in time (Belk 2013).

II. Current Solutions

To help users manage and filter through digital clutter, "memory" features and apps have surfaced, allowing users to view their memories in a meaningful form, such as a timeline or summary, rather than a random collection of pictures or links. Some existing solutions and their limitations are as follows:

Spotify Wrapped by Spotify

Spotify Wrapped is a personalised year-end experience that shows users their top songs, artists, and genres of the year, as well as fun facts and insights about their listening habits. However it has some limitations such as that it only uses the sense of hearing to stimulate memory and does not include other sensory information, such as sight, which can be just as important for memory recall. The wrap is also only based on Spotify listening data, so it does not include music listened to on other platforms hence it will not give the user an accurate representation of their memories.

Instagram

Instagram is a social networking service that allows users to share photos and videos. While it does allow users to have a multisensory experience, it is not as user friendly as a dedicated memories app.

Google Memories

Google Memories utilises Google Photos to create scrapbooks, grouping together similar photos. An interesting feature developed by Google Memories is the Cinematic photo feature, which works by animating together several similar photos. Google Memories, however, only works with photos and videos, and does not integrate music in its scrapbooks.

Our Solution

In general, we observed that while there are solutions which involve specific types of data, such as photos and videos (Instagram), and music (Spotify), there are none which integrate both together. Our solution, ScrapStitch is a mobile application that aims to help people preserve their memories effectively by combining social media posts, music, photos and journaling into one platform, allowing them to revisit their past using multiple senses (hearing and visual). The app will also encourage mindful memory creation, by prompting the user at random periods in the day to take a candid photo and write an entry, encouraging the user to experience and remember the mundane, day-to-day moments..

III. Solution

ScrapStitch serves as a convenient way to help users to look back and revisit all of their posts and music preferences from multiple platforms. Our proposed mobile application will have the following features:

Music App and Social Media Syncing:

ScrapStitch's main feature is the ability to link popular social media platforms and music streaming applications of their choice to our app. Our scrap and stitch system will then automatically collate uploaded posts and data. The user can later select a time period to view, opening up a collection of pictures, videos, posts and music. This would allow users to view their memories all from one app, without having to go through each application to search through their social media feeds. The user can also manually add pictures, music and journal entries if they want, providing additional context to the data automatically obtained from other platforms.

Candid Moments:

Additionally, ScrapStitch encourages users to document small, day-to-day moments in their everyday lives through random notifications which our app will send out throughout the day. This prompts people to take photos at random points in their day to capture vulnerable and realistic moments. Alternatively, users can choose to journal that point in time through plain text. These candid moments will be presented alongside other content in a scrapbook, offering a blend of carefully composed photos and videos that capture significant events, as well as impromptu snapshots that encapsulate the everyday, spontaneous activities that constitute a substantial part of the user's life.

Summarised Wraps:

Finally, at the end of every month or any specific time frame the user may choose, our app will compile a set of photos, videos, captions, journaling entries and create an aesthetically pleasing video with the user's top five/ten songs that were most played in that month. This will create a sense of nostalgia, allowing users to reminisce memories and past moments in their life, while providing insights on the user's preferences and patterns. This video can be shared with friends and family, or saved into storage for the user to revisit at a later date..

IV. Target Users

Primary Users: Young adults (18-24) who listen to music and regularly take photos and videos. Photo/Video sharing apps, such as Snapchat and Instagram, are especially popular among this age group (Smith and Anderson, 2018). In addition, from a survey taken from the SUTD population, we inferred that most of these users in this age group also regularly use journaling, social media, music, and file storage apps to view their past photos, videos, posts, and music.

Secondary Users: Users who take pictures/videos or listen to music, but not both, including those with audio/visual disabilities, and users who do use photo/video sharing app and listen to music, but do so less often than young adults, such as users in the adult category (25-49).

Supplementary Users: Elderly (60+) who rarely/ occasionally take pictures/videos and listen to music using mobile apps, but might want to give the app a try.

Served Users: Friends and family of the user who may share his “monthly wrapped” or memories with them.

Negative Users: Users who neither listen to music nor take pictures/videos. Users who are not interested in viewing past memories. Non-social media users.

Based on the insights shared from the initial survey, and our primary and secondary users, we developed 4 user personas.

User Persona 1: Jane (Primary User)



Profile

Gender Woman
Job Student
Age 21

Bio: Jane is adventurous and loves travelling. She enjoys taking pictures and documenting her adventures overseas.

Personality:

- Adventurous
- Sociable
- Creative

Goals

- Create a compelling story of overseas trips
- View and share experience and fun moments

Frustrations

- Troublesome to manually compile images into a collage

User Persona 2: Jacob (Primary User)



Profile

Gender Man
Job Musician
Age 23

Bio: Jacob is a budding musician who enjoys gigging and performing. He regularly seeks insights and inspiration to enhance his brand and music.

Personality:

- Passionate
- Introspective
- Free Spirited

Goals

- Enhance his personal brand and identity
- Seek inspiration for his music

Frustrations

- Finds it difficult to find insights from the vast amount of music he listens to

User Persona 3: Jonathan (Secondary User)



Profile

Gender Man
Job Sales
Age 25

Bio: Jonathan is a busy salesman who works in a 9-5 job. He tries to keep a healthy work-life balance, exercising and playing sports on the side.

Personality:

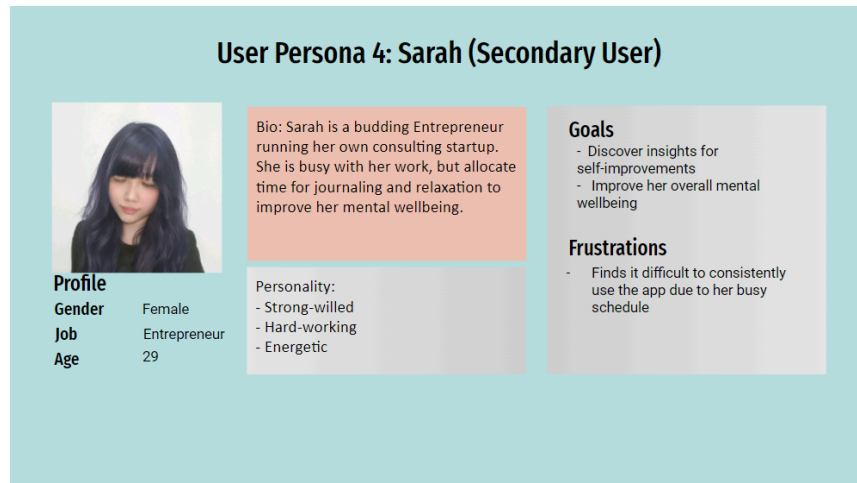
- Relaxed
- Hard-working
- Introverted

Goals

- Uncover insights of his music and photo preferences
- Revisit enjoyable memories

Frustrations

- Struggles to find time to take many photos due to his work.

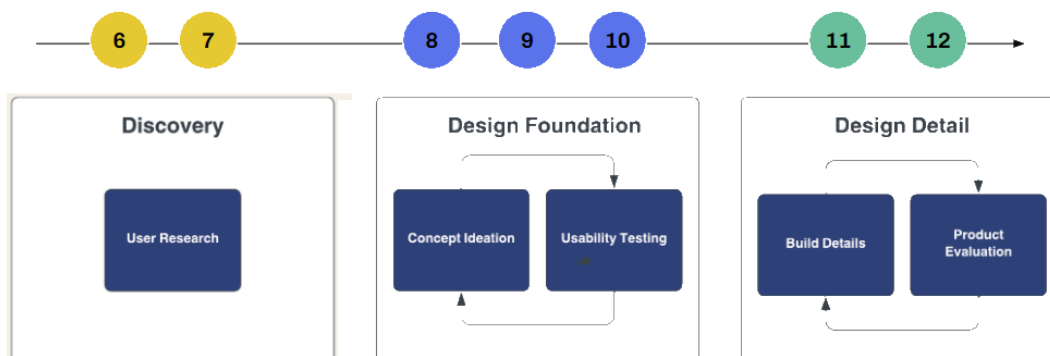


V. Timeline

Our timeline can be divided into three stages. These stages are:

- 1) [Observation](#) (Discovery) (W6-7). During this stage, we will focus on observing user behaviour patterns, understanding their mental models of our app, and identifying key insights that would help with our design.
- 2) [Design Foundation](#) (W8-10). Using these insights, we will iterate through different design ideas, following up with usability testing to validate our ideas.
- 3) [Build Design](#) (W11-12) Once we have designed a strong design foundation, we will continue building on it, polishing features and adding details.

Week



VI. Prototyping Methodology

During the Design Foundation stage, we used Figma to create lo-fi prototypes of apps. During each prototyping phase, the team will meet to discuss any important points and observations that would be useful before prototyping. Afterwards, the prototyping phase began and each member drafted a lo-fi

prototype. Afterwards, the prototypes were shown around and the team discussed and combined ideas from the prototypes into a full lofi prototype. This prototype was later used to test the viability of different design ideas (See [User Testing](#)). For our hi-fi prototype, a React app was created to simulate the functionality of the app as closely as possible.

VII. User Testing Methodology

Our user testing plan involved two types of tests: Qualitative and Quantitative tests. Both user tests involved the facilitator asking the user to do several tasks (eg. uploading an entry). However, both tests differ in purpose and were used at different stages of the project

Design Foundation Testing

Qualitative testing was used mostly during our Design Foundation stage after each prototyping iteration. During the test, there would be gathering insights, anecdotes, and problems discovered when using the app. This information was used to guide future prototyping iterations. In total, two iterations of prototyping and testing were done in this phase.

During our first iteration, we presented the user a list of tasks (view Wraps, add memory, view memories, view settings) to perform in sequence. We then noted down any unexpected interactions, and any tasks which they struggle at performing. After all the tasks are complete, we then asked the user to give a quick evaluation of the app, by rating the graphic design and user-friendliness on a scale of 1-5, and elaborating on what they did and did not like. We then tried to find out the user's interpretation of several design features, such as the colouring of the calendar entries, and the purpose of the tree in the home page, to check if they matched the intention of the design. Lastly, asked if they would personally use the app, and if so, what features they would use, to understand which parts of the app would likely see use. For the next iterations, we modified the list of tasks based on the new prototype, and we implemented A/B testing when evaluating which design decisions, such as the layout and behaviour of a page.

Design Detail Testing:

Towards the end of the project, our testing will focus on evaluating our product. Similar to testing in the design foundation phase, we introduced different tasks for the user to perform. However, our focus was on measuring as objectively as possible the usability of the app. Thus, we looked at more quantitative metrics. The metric are as follows:

- 1) Learnability - % of tasks completed successfully on first try
- 2) Efficiency - Number of taps or moves required to complete a task
- 3) Errors - Number of errors
- 4) Satisfaction - Subjective satisfaction from the user rated on a scale of 1-5.

In addition, we used Heuristic Analysis by evaluating if the design follows Jakob Nielson's Ten Usability Heuristics (See Appendix C for further details). By comparing these metrics, these tests were

justify different design decisions as the details and final features of the product. The evaluation can be found under [Design Evaluation](#).

VIII. Observation/Data Gathering

Upon more research, a clear relationship between multisensory experiences and memories is determined. Multisensory experiences act as a tool to improve one's ability to better remember situations/objects and recall those memories by binding the sensory cues with said memories.

As mentioned above, young adults spend most of their lives with a kind of mobile device in their hands. Hence, a survey with 31 responses was conducted to better understand our target audiences' phone application usage with respect to memories. The survey found that 96.8% of the respondents take pictures/videos (fig. 1) and listen to music (fig. 2) using their phones at least once a month. In addition, 90% of the respondents will revisit the music they used to listen to within a year's time (fig. 3), while 96.8% respondents will look back at their past pictures/videos/posts at least once every year (fig. 4). The few main reasons for doing so is to find memories to share with others and to reminisce and invoke memories (fig 5). Hence as pictures, videos and music play a big part of reliving and preserving memories for our target audiences, a closer inspection of the features in the applications used is done.

Music applications:

Firstly, music applications have features to summarise a user's yearly music listening pattern. An example of such a feature is Spotify Wrapped by Spotify. The feature creates lists such as User's Top Artist, Top Genre and Top 5 Songs. While features such as Spotify Wrapped does bring about memories and nostalgia, it only uses one of the 5 senses, hearing, to do so. Hence it is not the most effective method to relive a user's memories.

Journaling applications:

Next, there are journaling applications such as Notion, Phone Notes for users to input their memories and thoughts for them to look back at in the future. However, similarly to the music applications, journaling apps only use sight to relive and preserve the user's memory. Therefore, it is not effective when preserving and reliving memories.

Social media applications:

Lastly, social media applications that are commonly used by our target audience, allows users to post images and videos and include music with the post. One such example of this is Instagram. Although this may allow users to experience the multisensory experiences (hearing and sight) mentioned before, this solution requires the users to manually upload their photos and select the music to fit the theme of the post. This would mean that these applications are not as user-friendly for reliving and preserving the user's memories.

As most current solutions do not incorporate multisensory experiences into their applications and could be more user-friendly, our proposed solution will be able to tackle the issues mentioned.

IX. Prototype

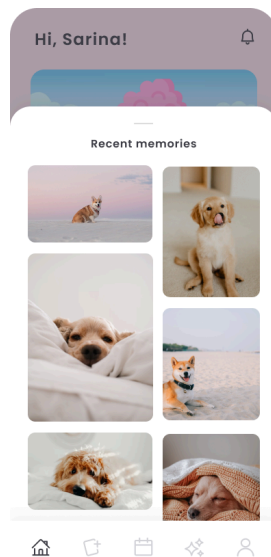
Based on the insights gained during the [observation phase](#), we identified a list of features which the design would need to implement. After creating our first two prototypes, we underwent a round of user testing, using the methodology in [Design Foundation Testing](#). The tests were held online via Google Meets calls so that the user's interactions and interview could be recorded for future analysis if necessary. The results are detailed in Appendix B. After many rounds of testing, we were able to identify key problems and solutions, which we implemented in the final Prototype. The prototype features and changes are detailed below.

Main Features

Music App and Social Media Syncing

With this feature, ScrapStitch will act as a one-stop platform for users to preserve and relive all the memories they have recorded in the past.

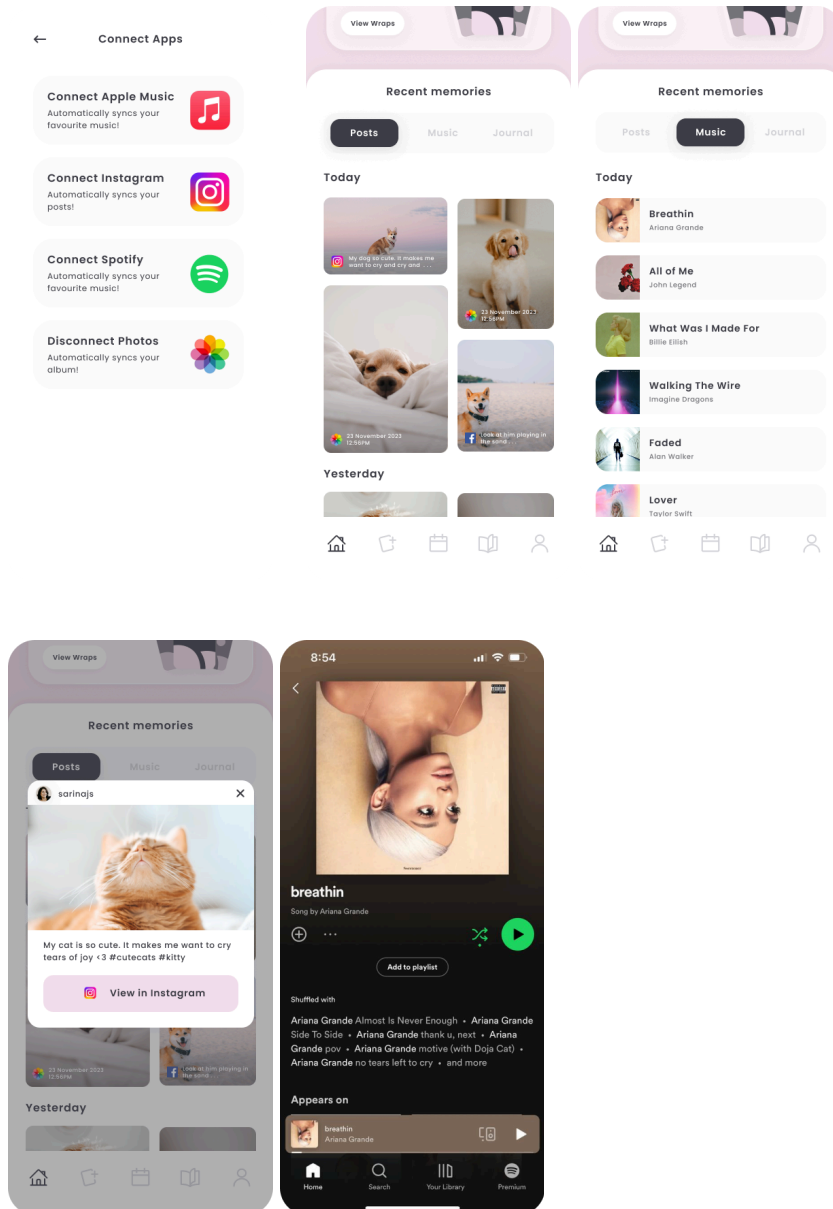
Prototype 2:



In Prototypes 1 & 2, this feature was only meant to help ScrapStitch preserve users' app data, where users will only be able relive their memories by viewing each individual post, image and the song but unable to interact with them. However, as users may open up the other apps to view the specific content after using ScrapStitch, this process will create a break in the flow of reliving memories where users are more unlikely to view the full memory behind a post/song. Therefore, to create a more seamless

connection between apps for the users to , buttons such as “View in Instagram” will bring the users to the post, image or song they are viewing at the moment.

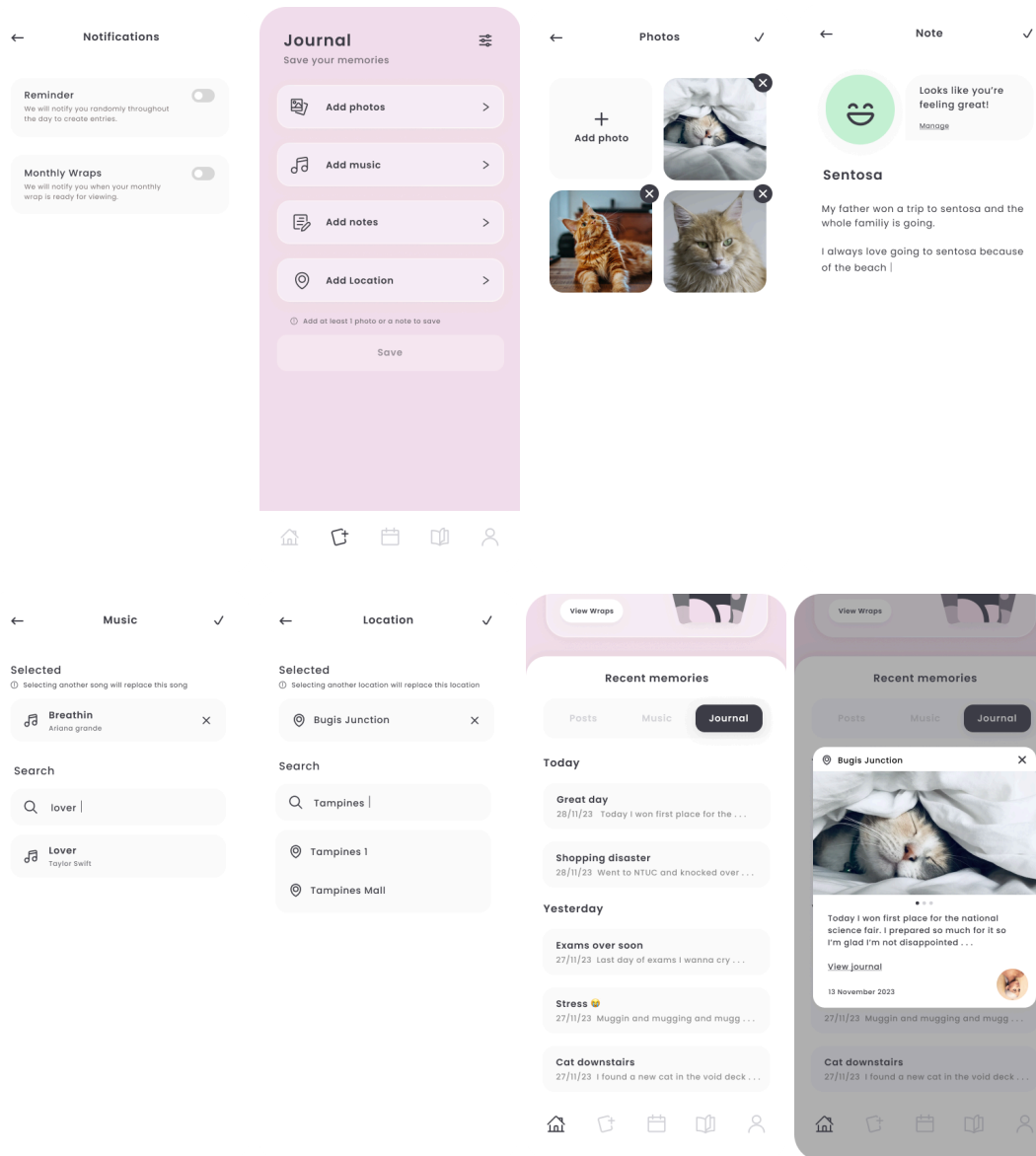
Final Prototype:



In the Final Prototype, users will be able to choose which apps they wish to sync ScrapStitch to (top left). Upon syncing up the apps, posts from social media apps, images and videos from photo apps and songs from music apps will automatically be added to ScrapStitch (2 examples to the top right). Users will be able click into each individual post or music, where they will be brought to the respective apps to relive their memories (bottom).

For cases where users do not wish to automatically sync their other apps’ data to ScrapStitch, they may still manually upload their respective data when using the other features of ScrapStitch.

Candid Moments (Notification + Journal Entry Pages)



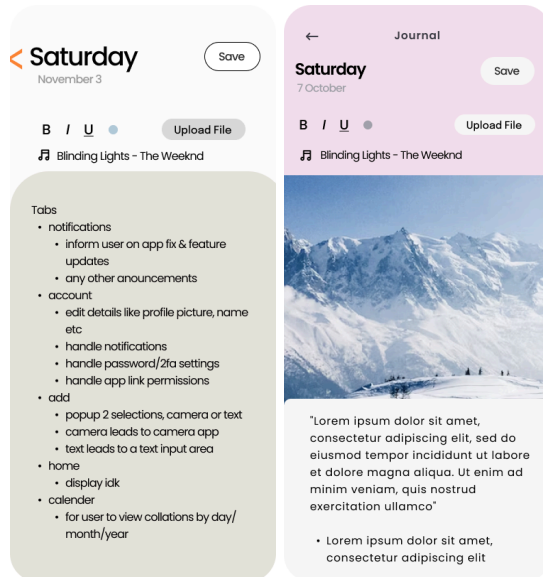
We introduced a random notifications feature to encourage users to write their journal entry in unexpected moments of the day. Users can choose to turn that feature on or off in the notifications setting page as shown in the first image. This helps users preserve memories of those unexpected moments in their lives and be able to look back at them as when they wish to.

Upon receiving a notification, the user can click on it and be directed to the add journal page as shown in the second image. This is where they can add photos, notes, music and location, image 3 shows the add photo page where they can upload multiple photos, image 4 shows the add note page where they can add a note with auto ai prediction of their mood, image 5 shows the add music page where they can

search for songs and add it, while image 6 shows the add location page where they can search for locations and add it.

After adding the journal, they can view it in recent memories in the home page as shown in image 7. When they click on the journal, there will be a popup modal which shows a summary of the journal content as shown in image 8. The modal has the location at the top, images and notes in the middle, date at the bottom left and music image at the bottom right. There is also a view journal button which brings them to view the full content of the journal and also allows them to edit it.

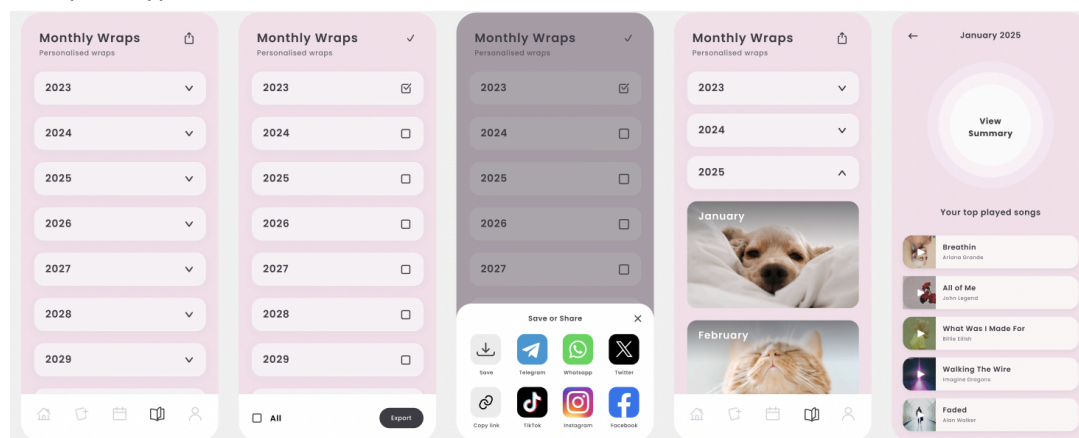
Prototype 1 & 2 respectively:



In prototypes 1 and 2, all the adding functions were crammed onto one page. Upon user testing, the feedback was that most of them did not know how to add music or location as the buttons were not intuitive. Hence, we decided to separate it into different screens to make it more intuitive.

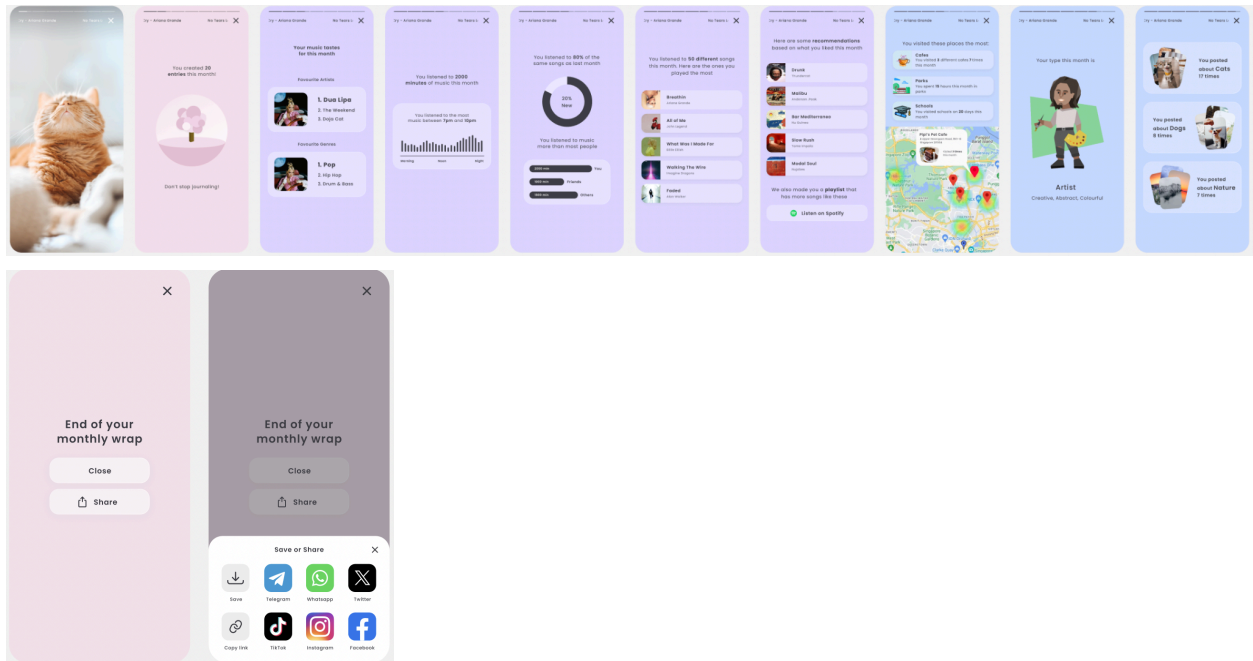
Summarised Wraps (Wraps Page)

Final prototype:



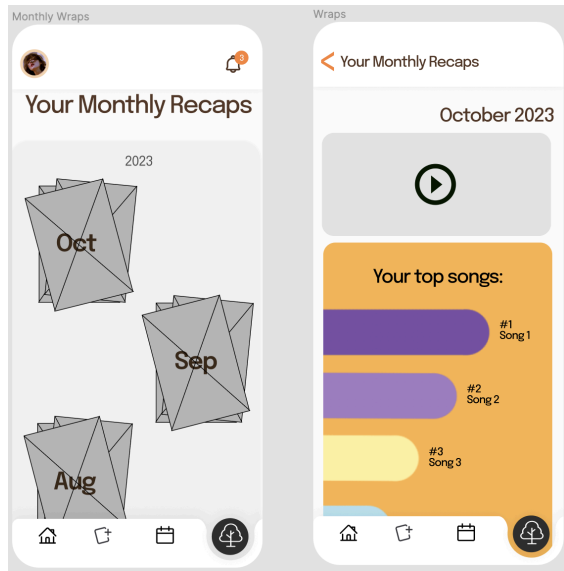
The Summarised Wraps page is a lookbook where users can see and recall their journey throughout the month. The page is arranged neatly in a way that makes users able to navigate with ease. Upon clicking on the Summarised Wraps Icon on the NavBar, users are shown the different years (2020, 2021, 2022, 2023), depending on how long they have been using the application for. A notable function for this page is that users can select the year of summarised wrap and save it to their mobile device or share it to any social media platform they want. If the user just wants to view their past recollection, they can also just select on the year and a drop down of the months collated in that year will be shown. Users can then choose the month they wish to revisit and start reliving the moments.

Final prototype (Monthly wrap):



They will be shown their top songs that they have listened to that month and also a summary of journal entries, and pictures written and taken that month in the form of a summary video. One of the unique aspects of our summary wraps is the way the wrap is displayed to the users. The wrap is shown in individual pages in a very dynamic way so that users can reflect on the month. The Summarised Wrap does not just plainly show the top moments of the month, it showcases it in an interactive way where it analyses the type of data that was fed into it. For example, based on the songs the user has listened to the most in that month, the app will recommend similar songs. Another analytical feature the Wraps does is take into consideration the places, pictures, songs, and journal entries and categorises the type of person that the user is. For example, as shown here in the screenshot, the Artist is a creative, abstract, and colourful person. Finally, this monthly wrap also has the option to be shared to anyone in particular like close friends and family or can be shared on a social media platform.

Prototype 1 & 2:

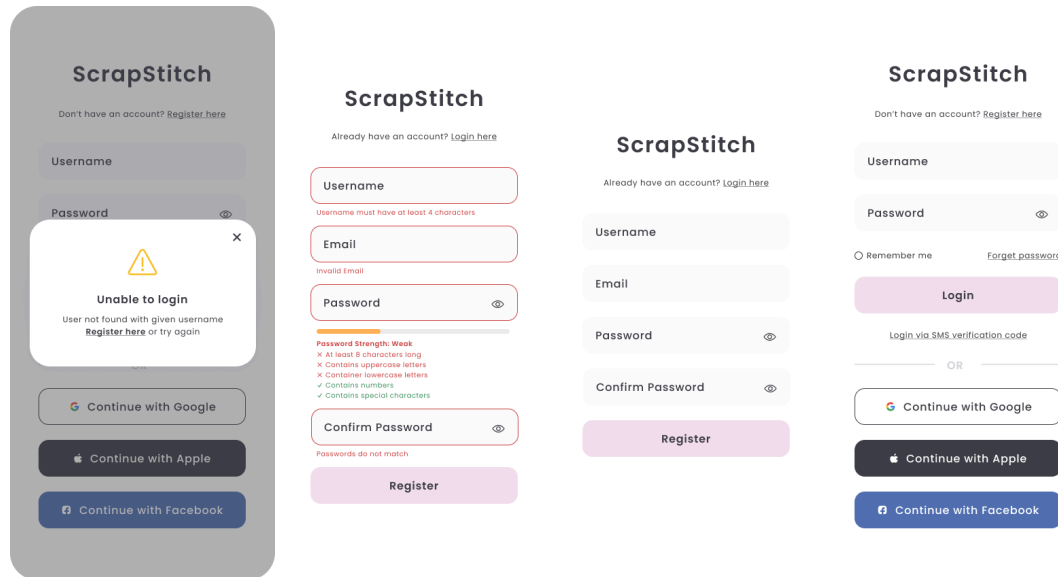


In Prototypes 1 and 2 the Monthly Wrap was simply lacking in a uniform aesthetic that was both pleasing to the eye and easy for users to navigate. Upon user testing and multiple trials and errors, we came to the conclusion and decision to change the colour scheme and the formatting of the monthly wrap. The idea for the information to be had was there but the execution was lacking at this stage.

Side features

Login Page

Final Prototype:



Existing users can log in using their usernames. New users will be able to create an account using their email. They can also choose to make use of Google, Apple and Facebook's sign in services to login and register.

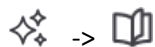
Navigation Bar

Final Prototype:



A navigation bar will be shown at the bottom of each main page of ScrapStitch for easy navigation around the app. From left to right, the icons for the respective pages are: Home Page, Journal Entry Page, Calendar Page, Wraps Page and Account Page.

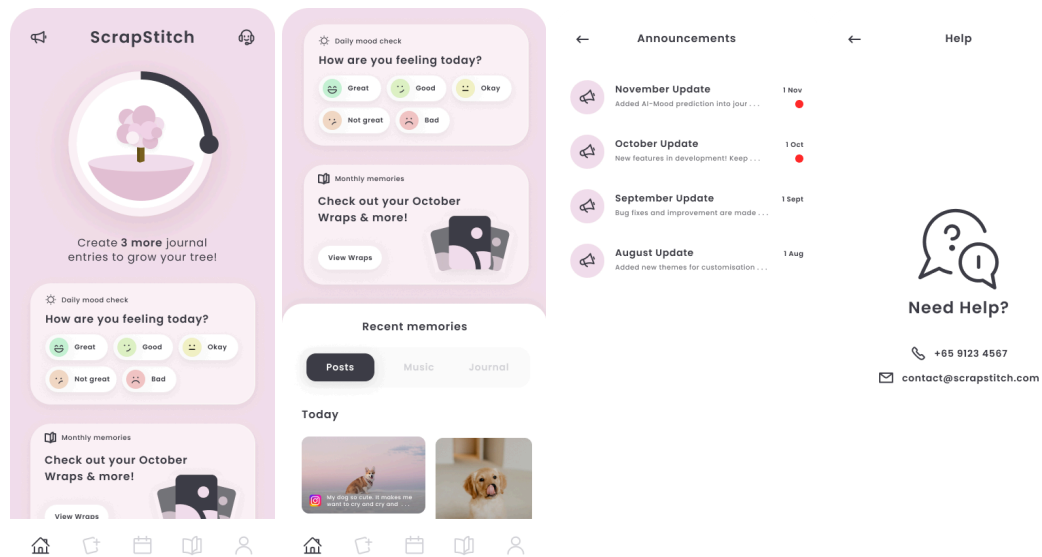
Iterations of Wraps Page icon:



The icon for the Wraps Page has been changed from a sparkle to a book as users did not find the previous sparkle icon to be intuitive.

Home Page

Final Prototype:



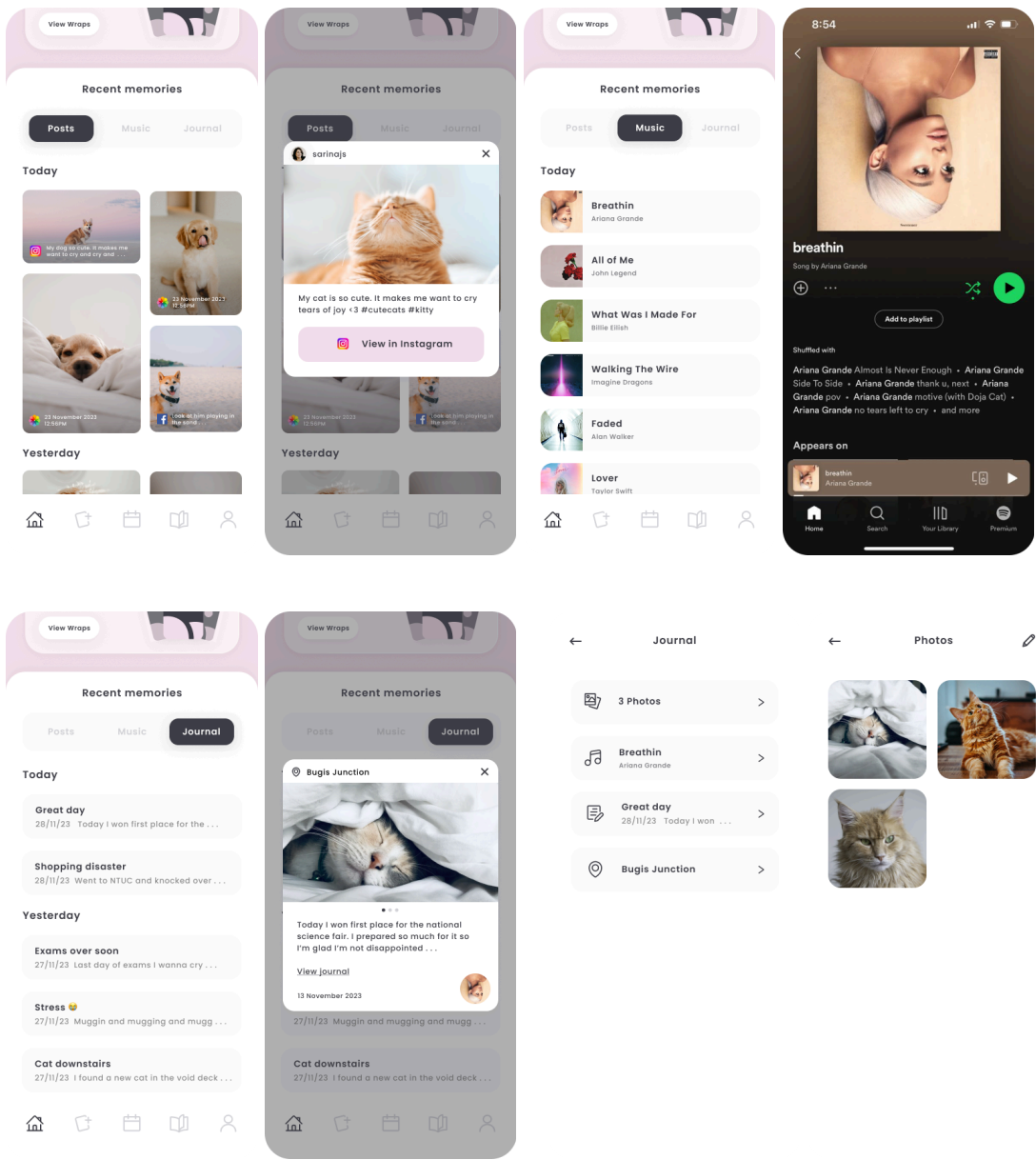
The Home Page acts as a summary and welcome page for users when they open up ScrapStitch. They will be able to view announcements, after pressing the speaker icon at the top left hand corner, such as the latest updates and newly available monthly wraps. Customer service help will be available as well after clicking into the top right hand corner.

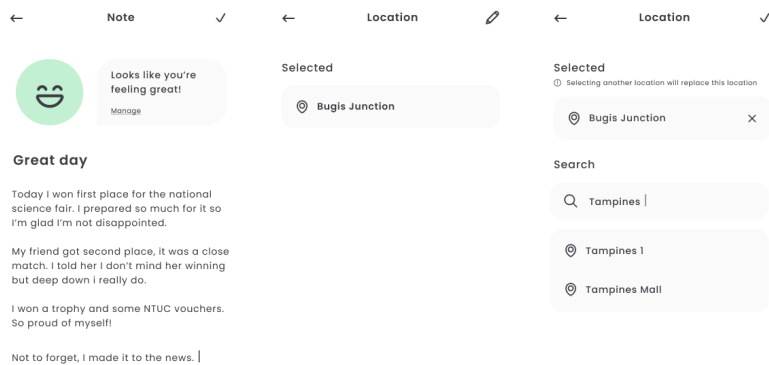
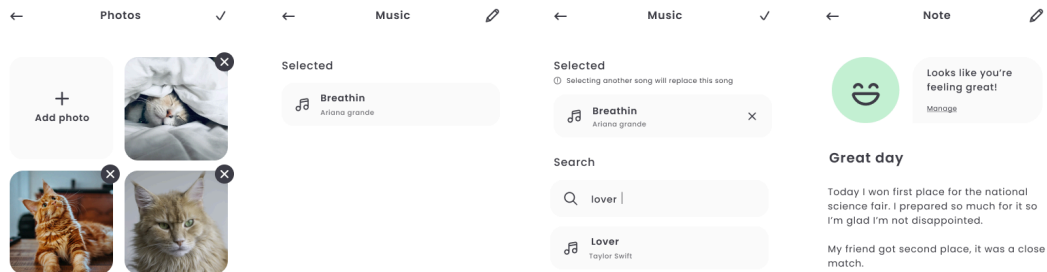
To incentivise users to use more of ScrapStitch's Journal Entry feature, a tree at the top of the page will grow as more entries are created. A text to inform users of the number entries needed to grow the tree will be available to motivate them as well.

As mood plays a big part in reliving memories, a daily mood checker on the Home Page will remind users to input how they are feeling on a certain day. The mood data can be found later in the Calendar Page for users to look back on as a way to preserve and relive memories.

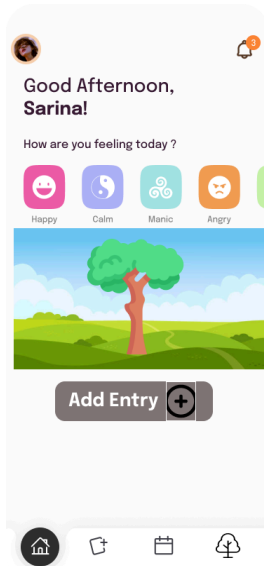
As Summarised Wraps are one of the main features of ScrapStitch, users will be prompted to view their monthly memories by clicking on the "View Wraps" button or the book icon on the navigation bar.

Final Prototype (Recent Memories):





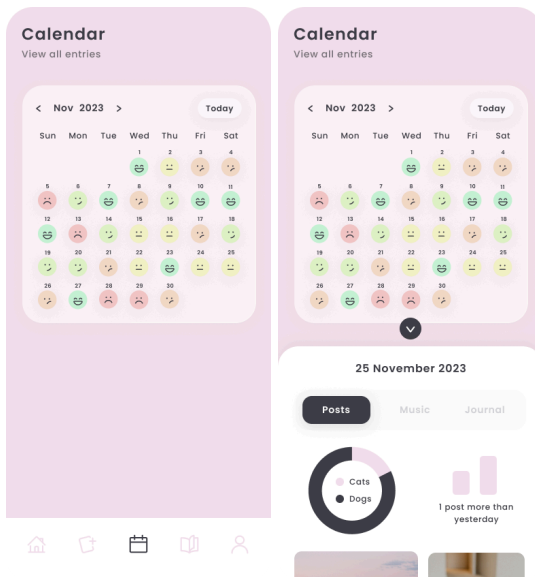
Prototype 1:

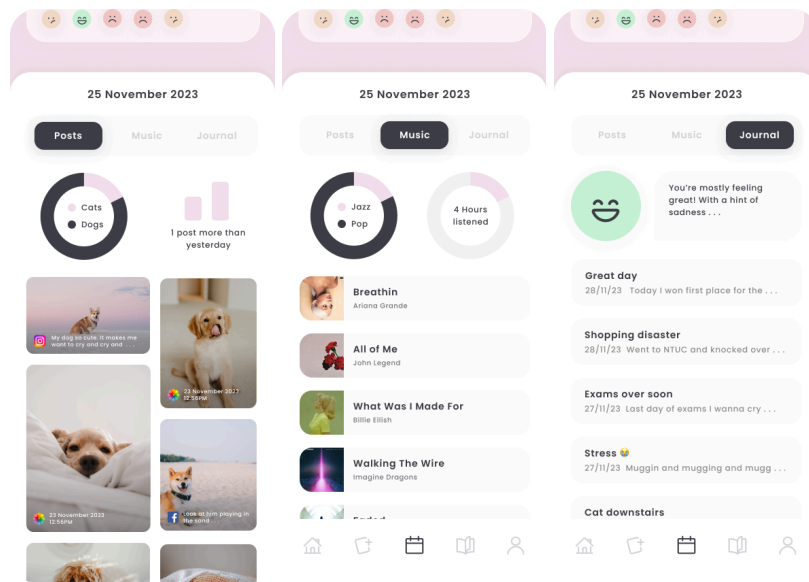


User's recent memories of their posts, music and journal entries will be available for viewing as well in the Final Prototype. This feature is widely enjoyed by our user tester as it allows users to immediately see all of their most recent memories after a simple swipe up on the home page.

Calendar Page

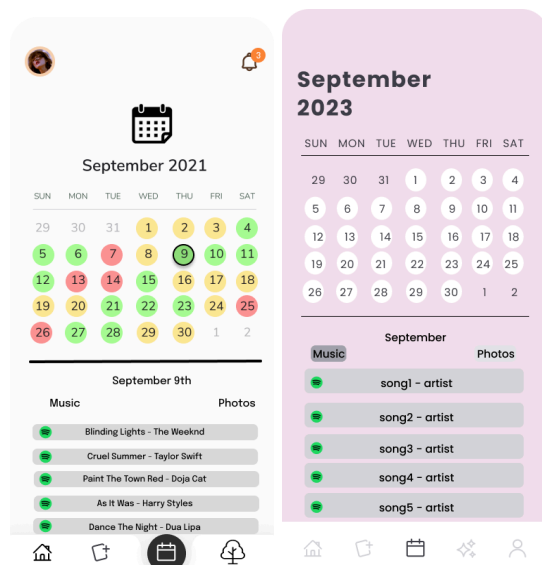
Final Prototype:





The Calendar Page acts as a database for the user's daily memories. In the Final Prototype, upon entering the page, the user's daily mood, which they have chosen on the Home Page previously, will be displayed for their respective dates. By clicking on any one of the dates on the calendar, statistics of the user's type of posts, genre of music and mood data will be summarised and displayed on the bottom section together with all the post, music and journal entries in the respective tabs. This section provides daily fun facts which are different from the monthly or yearly fun facts that are found in other apps mentioned before.

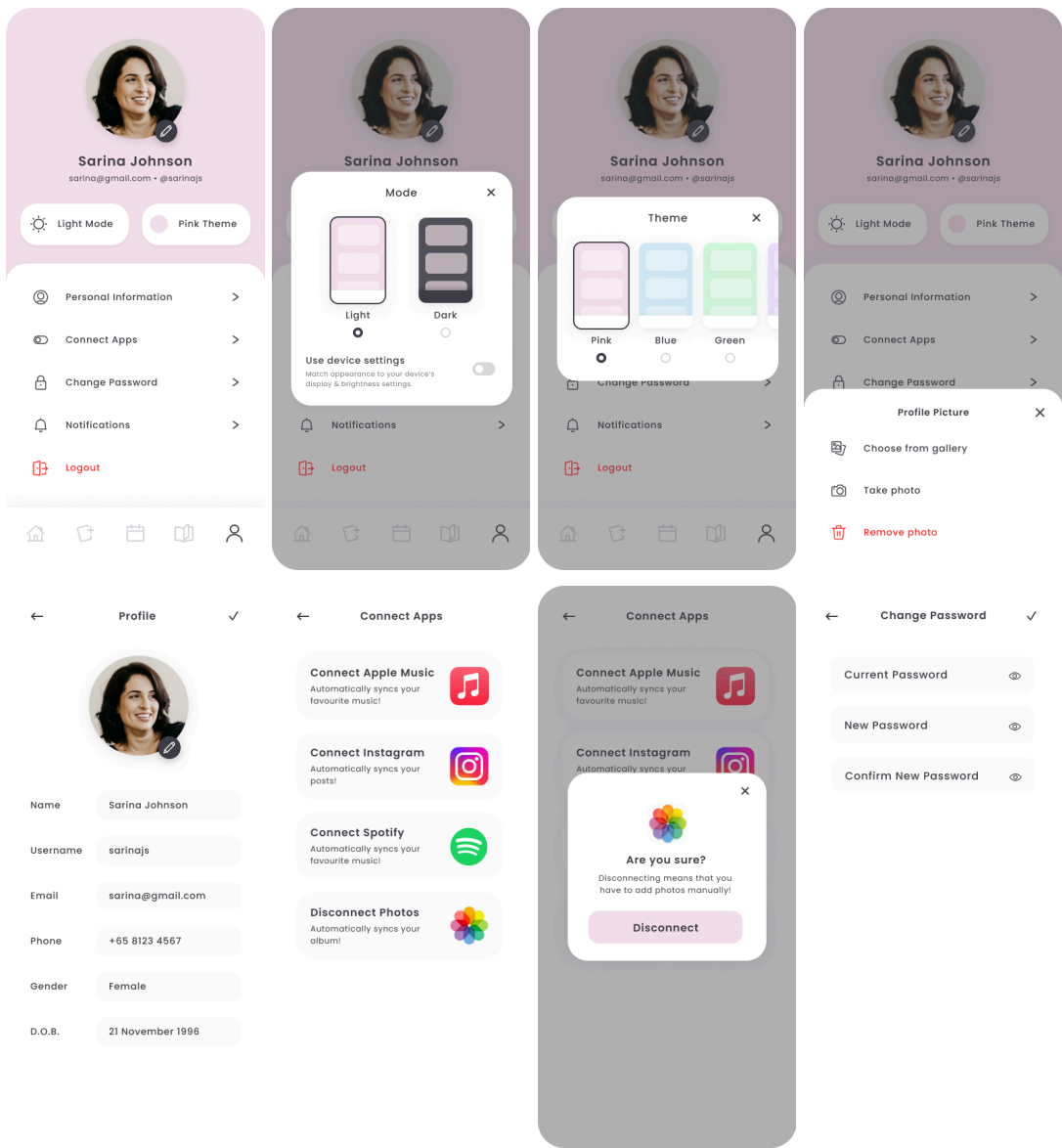
Prototype 1 & 2 respectively:

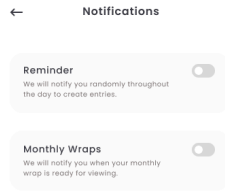


In prototypes 1 and 2, the feedback from user testing was that it is not intuitive to scroll down to the bottom section after selecting a date. Hence a scroll prompt was added in the form of an arrow down button. This prompts users to scroll down or click on the button itself which will both bring them to the bottom section.

Account Page

Final Prototype:





The Account page includes many features such as allowing users to view and change their account's information, colour theme of the app, manage the apps they have synced to as well as the types of notification they wish to receive.

Colour themes and light/dark modes have been added to the Final Prototypes as there was feedback that users are more likely to use ScrapStitch at night. The changing colour themes will keep the app visually fresh and cater to user's preferences. The colour of the tree on the Home Page will change together with the theme as well.

X. Scenarios

Scenario 1 (Jane)

Jane is an adventurous university student who loves travelling and taking pictures of her journeys. She has been planning to take a gap year from her studies in order to travel more. She takes many pictures during her travels and posts a lot of them to her instagram and other social media platforms. As she is planning her travels she is also looking back on her previous trips and realises that often her pictures are lost in her photo album and thinks it is difficult to find important pictures from certain moments in her photo album with all the photos she has taken. She also remembers that she does not have the time to manually compile her photos into a collage during travelling and forgets to do it after her trips. She gets worried that her memories from this big trip are going to get lost the same way as her other trips.

To make collecting her memories easier Jane downloads ScrapStitch. During her year of travel she takes pictures of the places she goes to and after every month she gets an automatically made wrap from ScrapStitch showing the highlights of her trips. These wraps are saved in her app so she can look at them after her trip and she can also share them to her friends. Jane has also enabled the notification feature on the app and gets reminded to take a picture randomly by the app everyday and write her thoughts

relating to this picture. This helps Jane capture also the moments of the trip that aren't the most exciting but just as important in making her remember the trip. Jane has not written a journal this extensive before. During her trip she is listening to a lot of different music. This music gets combined with the pictures she takes so when she looks back she can get a multisensory memory of the time she was travelling.

Now Jane can take all the pictures she wants during her travels and does not have to worry about making them into a cohesive collage. She also has a database to go back to and easily find the moments she wants to remember

Scenario 2 (Jacob)

Jacob is an aspiring musician who loves getting inspired by the world around him and the music he listens to daily. Jacob has recently began gigging and performing around his local area and wants to enhance his personal brand and identity. He struggles to find insight from the vast amount of music he listens to.

Downloading ScrapStitch Jacob is able to see what music he listens to the most on a monthly basis and based on his listening habits the app is able to give him suggestions on what new music he could listen to. This helps him get inspired in making new music. The app also gives him statistics about his listening and he is able to find the common theme in his music and therefore makes it easier for him to find his brand identity.

Scenario 3 (Jonathan)

Jonathan is a busy nine to five salesman who tries to keep a healthy work life balance by exercising and winding down after work. Jonathan however struggles to make memories from his everyday life. Also he would like to have an easier way to revisit his memories from past experiences.

Downloading ScrapStitch he is able to capture his everyday life and write about his normal day to day activities more without even thinking about it since the app gives him daily reminders to write down his thoughts. He is able to track his writings through the calendar page. He is also able to go to the ScrapStitch wraps and relive his past memories in a short slideshow while he is winding down from work.

Scenario 4 (Sarah)

Sarah is a very busy entrepreneur running her own consulting startup. She is often busy with work but likes to allocate time for journaling and relaxing to improve her mental health and keep her mind clear. She is very interested in self-improvement and a big advocate for mental health related research. She knows that screens and social media make her mental wellbeing worse and enjoys writing journals on paper.

Writing her journal takes a large amount of time since she has to reflect on the whole day at the end of it and she struggles to remember everything. She finds it hard to allocate time for writing her journal as her startup requires more of her involvement. She downloads ScrapStitch in order to write down her

thoughts during her busy work day. She uses ScrapStitch's notifications to get reminded to write a journal entry during her busy workday. She however writes just a short paragraph without thinking about it too much. Then when she gets back home and starts writing her proper journal entry she can look back at her thoughts and pictures from that day and can get a more descriptive entry.

XI. Design Evaluation

We conducted our testing based on the plan highlighted in [User Testing Plan](#), analysing the number of errors encountered, and tasks completed on first try. We also calculated the minimum number of clicks required to complete each task (from the home page) to evaluate the efficiency of the app. The tasks tested are shown below, which were given to the user to perform in order.

- 1) Create a journal entry, adding a photo and song.
- 2) View Wraps.
- 3) View and edit an existing entry
- 4) View Settings
- 5) Update Theme

User Testing Metrics

Efficiency wise, most tasks could be completed in 1-3 clicks. Task 1 however required many clicks if the user wanted to add different components, such as music, photos, and text.

For learnability, except for Task 2 (66%) and Task 3 (66%), all the tasks were able to be performed successfully on the first try by all participants.

To measure subjective satisfaction, we asked the users to rate how satisfying it was to use the app on the scale of 1-5. The average score was 4.3.

Usability was evaluated using the System Usability Scale (TODO: add appendix). Each statement, related to the usability of the app, was given to the tester, who evaluated how much they agreed with the statement on a scale of 1-5. The results were collated into a score measuring the general usability of the app, with an average score of 43/50 (a higher score indicates higher usability). Refer to Appendix C for details.

Heuristic Evaluation

Our design was evaluated according to the 10 Usability Heuristics:

#1: Visibility of System Status:

Navigation bar icon is highlighted to indicate what tab the user is in and page titles are indicated when the user is viewing a journal entry or adding a new one. On adding an entry, the user is notified that he is successful.

#2: Match between System and Real World

The jargon used in the app is straightforward, with the exception of the term “wraps”, which is based off Spotify’s yearly wraps.

#3 User Control and Freedom:

Users can go back or exit any window by pressing the back button, either from the phone or the app.

#4 Consistency and Standards:

We maintained consistency with external standards by placing the settings/profile on the right most tab and Home on the left most tab. Internally, a Style Guide (Refer to Appendix E) was used to standardise the colour scheme and font.

#5 Error Prevention

Entries can be easily deleted and edited from the app

#6 Recognition rather than recall

Wraps and Memories can be accessed directly from the home page so the user does not need to remember which tab to navigate to.

#7 Flexibility and Efficiency

Tabs can be accessed directly from home if needed.

#8 Aesthetic and Minimalistic

App design is minimalistic and focused on conveying data without displaying too many elements.

#9 Help Users Recognize, Diagnose and Recover from Errors

Error messages are given in the case of login or syncing errors to help guide the user to solving the error.

#10 Help and Documentation

As it is, this app does not have help and documentation as it is relatively simple. However, this may be a point to improve on as more features are added.

XII. Reflection

This project provided the team an opportunity to learn about and practise the UI/UX design process. There were several insights gained throughout the project timeline.

Balancing Flexibility and Usability

Due to the nature of ScrapStitch, which takes in many different forms of data, such as photos, music, journal entries, location and emotion, it was challenging to decide on which functions of the app should be customizable and which should be fixed, such as the app colour scheme, and how the wraps are created. Having a customizable interface would help cater to users with different preferences, but having too much customizability would make it straining and difficult for the user to use the app.

The testing phase proved important when understanding the right balance between customizability and usability. When designing and testing our prototype, we discovered that different testers may have differing, and even contradicting opinions. Some like the modern, simple colour scheme of our first prototype, while some gave feedback that they felt it was too plain, and would prefer more calming colours. Thus, we added the option to customise the colour scheme and mode of the app. Meanwhile, the wraps function was well received by testers, and when asked about what they would like to add or remove, most users indicated that they liked the current pages. Therefore, we did not include any options to customise the wraps features. The key to finding a good balance between flexibility and usability was ultimately continuous testing and user observation, providing customizability when users may have a wide range of preferences, and removing unnecessary options when it does not significantly affect the user's experience..

Designing Data Displays

The key concept of ScrapStitch is the ability to store and view meaningful data in the form of memories. Throughout the Design Foundation phase, we explored many different ways to display photos, journal entries, and music to the user, which would provide a concise and meaningful story to the user. One feature which we changed a lot was the calendar view. We initially designed a prototype using green, red and yellow circles on each date, but the team was split on whether the colours should represent the number of entries, or the mood of the user on that day. While the principle we wanted to follow was to ensure our designs were consistent with other apps, testing revealed that this was not so simple to do. One user guessed that red meant more entries, while green meant less, as it was similar to Github's commit view. Several users associated green with good moods and red with bad moods. Others thought that green meant more entries as the app was encouraging storing more memories.

Our initial solution was to have a dropdown to filter by mood and number of entries. In the view filtered by the number of entries, the dates would have the same colour, but had different levels of saturation. In the mood view, the entries were coloured according to the mood-check box in our Home Page. This introduced greater clarity, but it was not a perfect solution, as users still struggled to identify which moods were happier, and which dates had more entries. In our final iteration, we made the change to scrap the number of entries' view and focus on mood instead, and our mood was also indicated by emojis, with a happy face for good and a sad face for bad. This removed any ambiguity and allowed users to immediately identify which days had happier moods. When designing and comparing different forms of data visualisation mediums, it is important to consider all possible conventions or standards, both by researching past user interfaces and observing the user's mental model, to ensure that the interface is clear and easily understandable.

Overall Project

As the theme of our project was memories, we all understood the importance it is in this day and age to preserve as many memories as possible without taking away the joy in the present moment. With thorough research, we believe we were able to find ways to tactfully tackle this issue in a way that caters to a large group of people and also in a simple and adaptable manner. One of the fundamental aspects of this project we had to understand apart from the problem itself is also the users. We had to dig deep and empathise with and understand their needs and desires in preserving memories. We conducted user research through surveys to gain valuable insights into their preferences, pain points, past experiences, and expectations. In conclusion, the overall project taught us that background research on the topic at hand and also on the users themselves were just as important as the designing aspect.

XIII. Conclusion

With the transition to a digital-centric world, and as the amount of data being preserved continues to grow, ScrapStitch offers a convenient, personalised and engaging platform for users to preserve, reminisce, and share their memories. The fusion of user insights, aesthetic considerations, and iterative design processes has resulted in a product that not only reflects the emotional significance of memory preservation but also strives to enhance the user experience. By integrating the users' visual and audio memories, spontaneous and planned, ScrapStitch empowers users to capture and cherish every precious moment.

As we conclude this report, the hope is that ScrapStitch not only meets but exceeds user expectations, becoming a seamless and integral part of their lives. Ultimately, the success of this project lies not only in the aesthetics and functionality of the user interface but in the positive impact it has on users' lives as they embark on a journey of preserving and reliving their most cherished memories.

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www.pewresearch.org/internet/2018/03/01/social-media-use-in-2018/.

Appendix A: Initial Survey Results

Supporting Figures

How often do you take pictures/videos on your phone?

31 responses

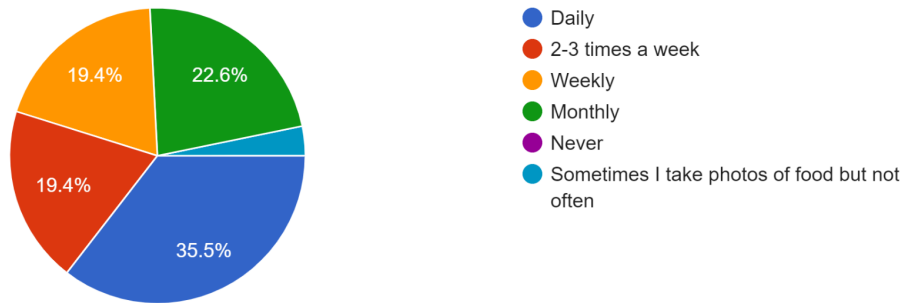


Fig. 1 How often do you take pictures/videos on your phone?

How often do you take pictures/videos on your phone?

31 responses

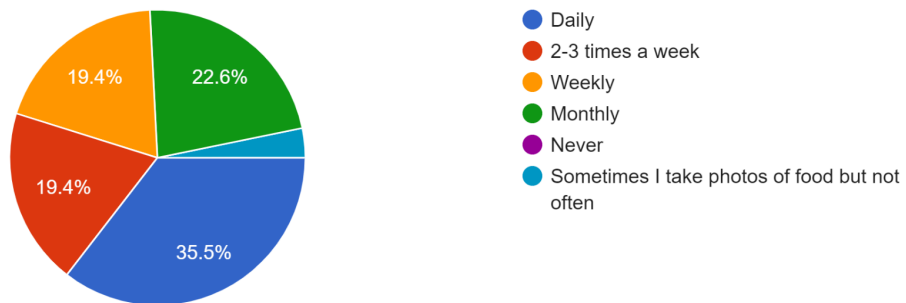


Fig. 2 How often do you listen to music?

How often do you revisit music you used to listen to?

30 responses

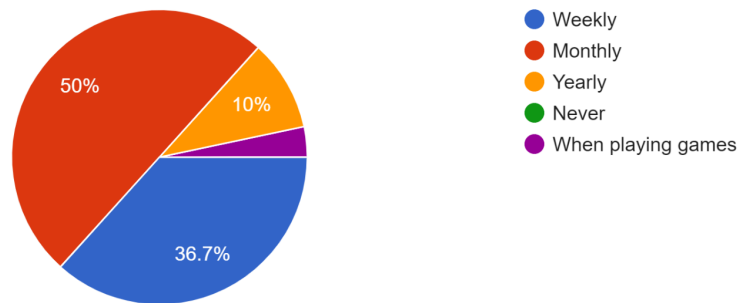


Fig. 3 How often do you revisit music you used to listen to?

How often do you look at your past pictures/videos/posts?

31 responses

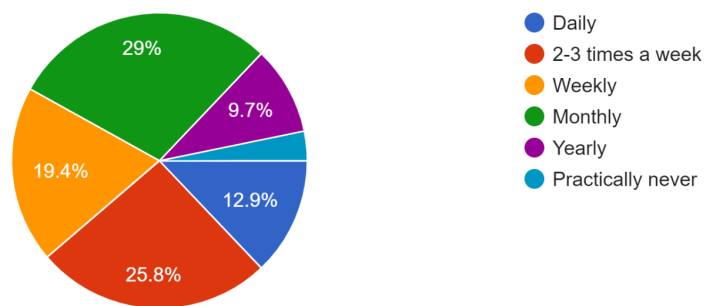


Fig. 4 How often do you look at your past pictures/videos/posts?

Why do you view past pictures/videos/ posts/music?

30 responses

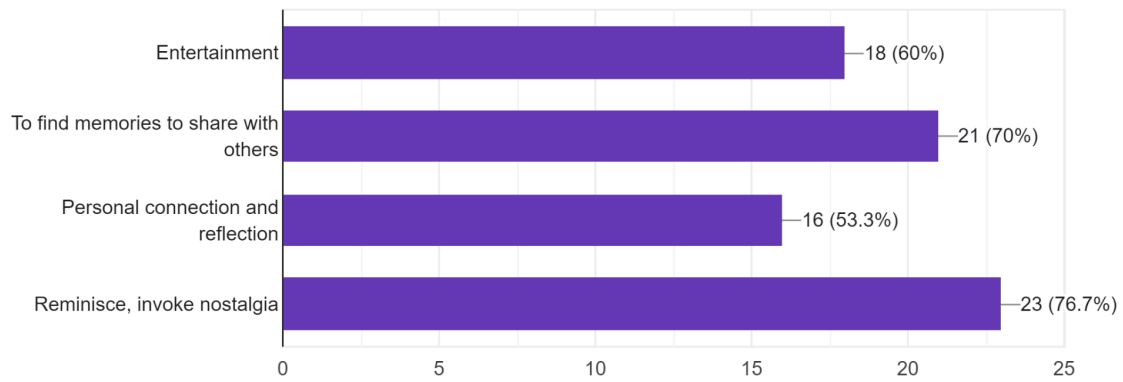
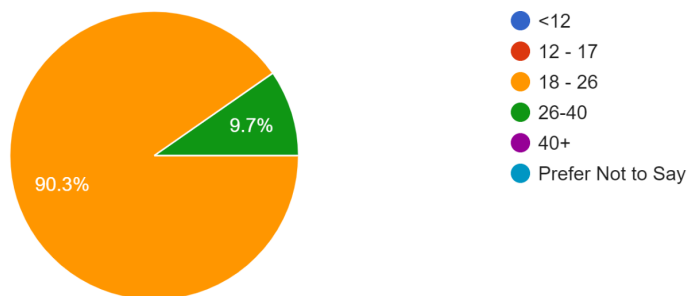


Fig. 5 Why do you view past pictures/videos/posts/music?

Survey Responses

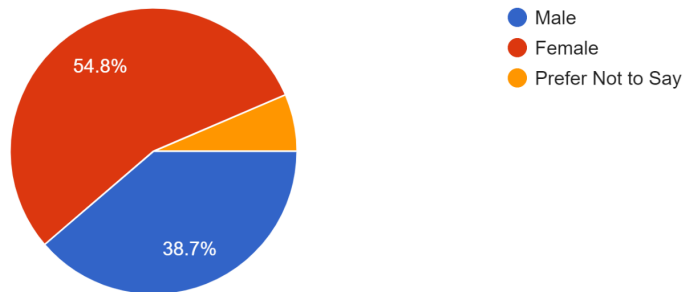
Age Group

31 responses



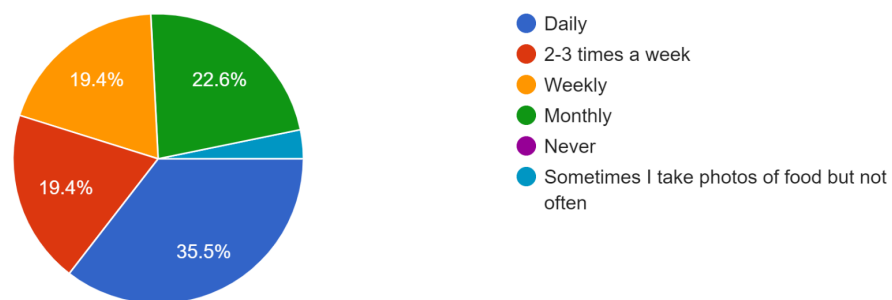
Gender

31 responses



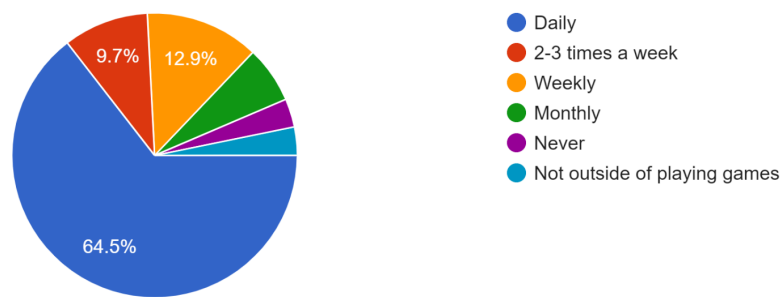
How often do you take pictures/videos on your phone?

31 responses



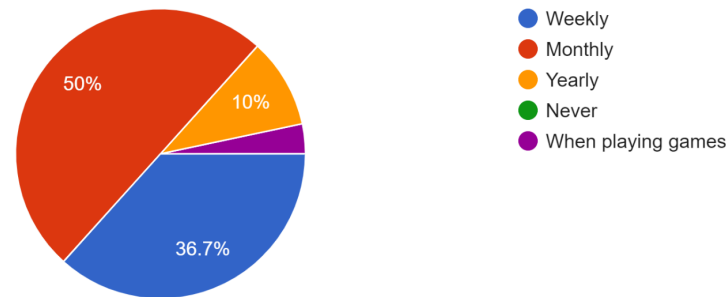
How often do you listen to music?

31 responses



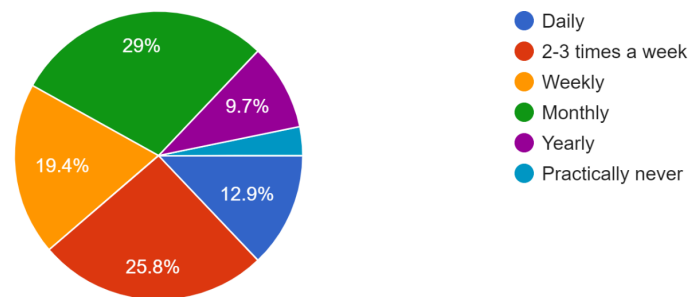
How often do you revisit music you used to listen to?

30 responses



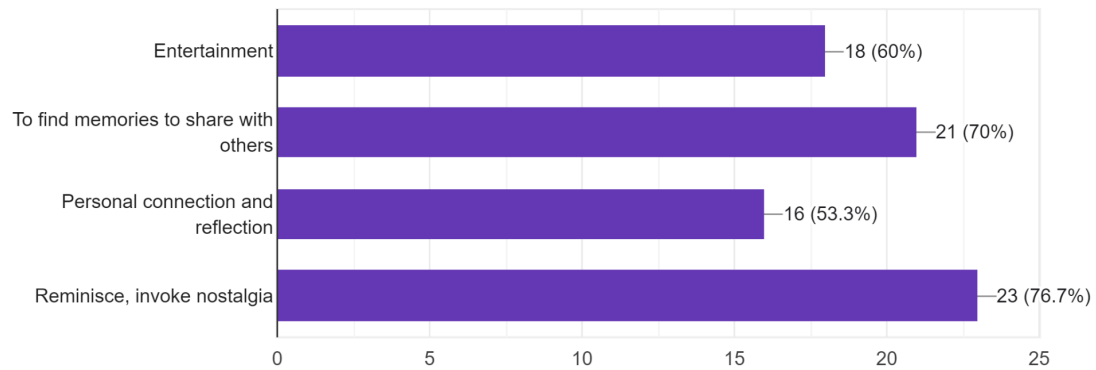
How often do you look at your past pictures/videos/posts?

31 responses



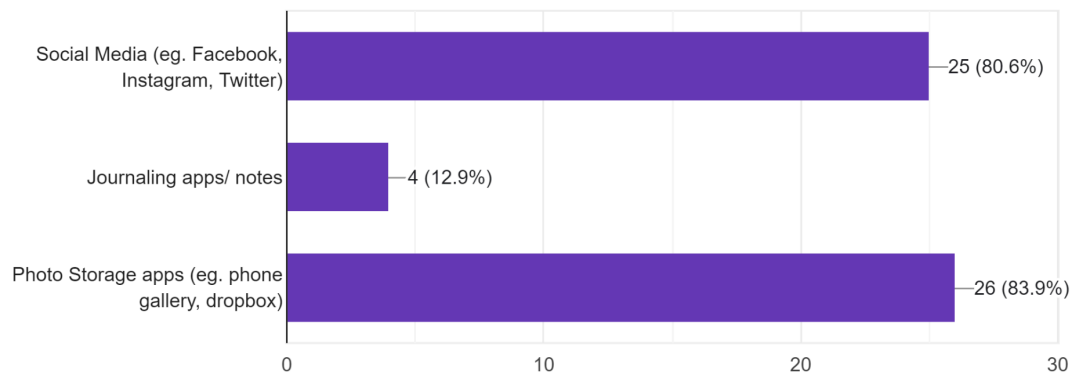
Why do you view past pictures/videos/ posts/music?

30 responses



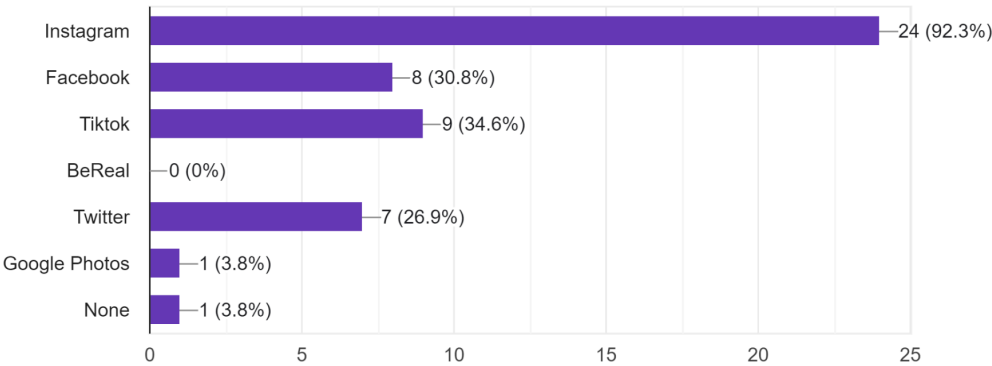
What apps do you use to view past pictures/videos/posts?

31 responses



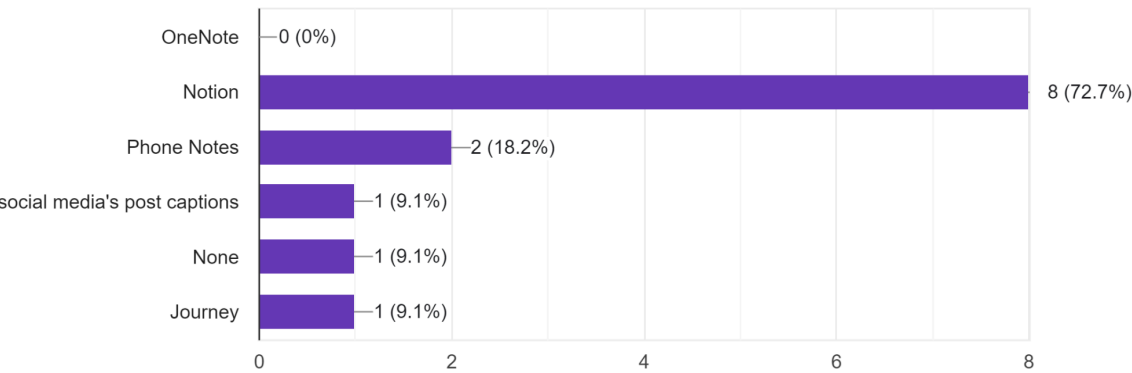
Which social media apps do you use to view past pictures/videos, if applicable?

26 responses



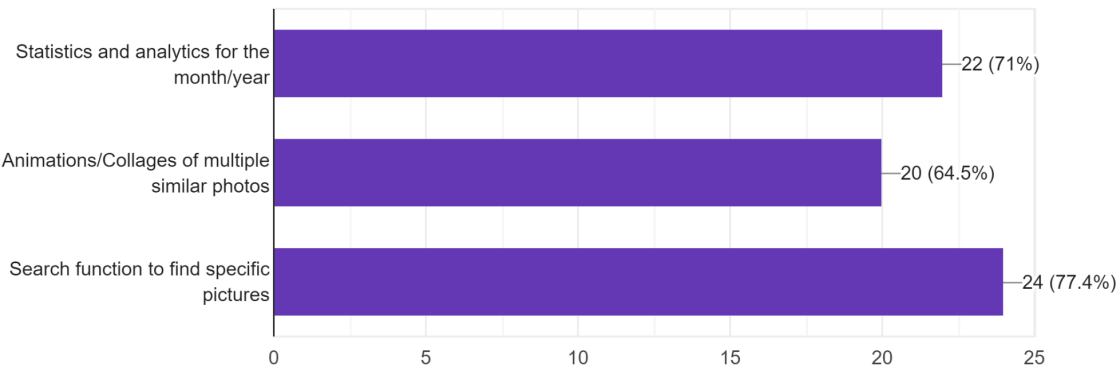
What journaling apps do you use to view past posts/photos/videos, if applicable?

11 responses



What features would you like to see for using an app to view past memories?

31 responses



Appendix B: User Testing Sessions

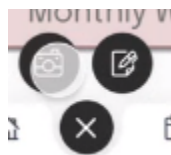
Testing Iteration 1

1. Brandon

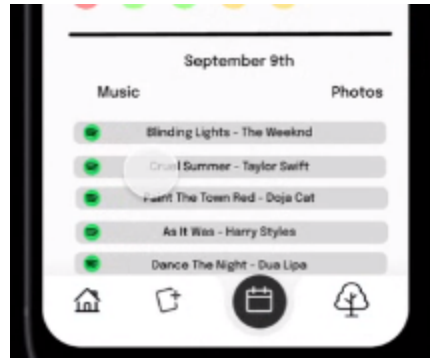
- Struggled to find the entries under the calendar, tried to click the tree and the wraps icon instead
User journey / overall thoughts
- Colour of circle in calendar is confusing
 - he thinks it's by volume of entry (red is more entry, green is less)
- Only managed to guess that wraps is under tree icon (maybe can change the icon to something else)
- Wraps portion can be more prominent (since its one of the feature, the first thing that he thought where the wraps would be would be clicking on the tree in the center, profile at the top left hand)
- Pretty cute, don't have to think too much, users use frequently, build a sort of familiarity with it, nice flow, have everything on the first page
- Would like a search bar for...

Survey questions

- On a scale of 1-5 how would you describe the aesthetic
 - 3.5
 - Why did you give that score?
 - White background generally nice, dark mode is nice (journaling is at night, in bed)
 - Too much colour for now
 - Nice cartoonish thing, become hi-fi will be great



- likes this



likes how you can see the music

used/chosen

- On a scale of 1-5 how easy was it to complete the tasks?
 - 4
 - Why did you give that score?
 - Wraps is a minus
- On a scale of 1-5 how user-friendly is the app?
 - 4
 - No comment
- Would you use the app? Why or Why Not?
 - Personally no. He doesn't write down stuff, has friends to talk and rant to
 - Would you use it if the app is linked another app? Yes. fun to know stats. Go overseas, post a lot of insta photos, good to know how many i posted
- What kinds of fun facts/ would you like to see in the monthly/yearly wraps
 - Mood board
 - Catered more towards music, suggest similar songs
 - Personality this month
 - This year did some different stuff, listen to a certain kind of music, we think you are a ... (hardworker, zen worker, gymbro, weeb, a certain artist's fan)
 - Generate something special (ask if they went to the concert, some fun fact about artist)

2. Jodi

- Struggled with finding where the wraps were, clicked on the tree instead.
- What do you think of the aesthetic?
 - Monthly recaps page vibes feel different from rest of the pages
 - Colour scheme should be same for all pages
 - Not sure what the colours in calendars mean, guess its mood, feel not appropriate in terms of colour
 - Monthly recaps icon not intuitive

- The rest not bad
- How easy was it to complete the tasks?
 - Overall quite intuitive except wraps
 - doesn't realise can click on settings button
- User friendliness
 - Generally okay, could show intermediate button actions
- Would you personally use the app?
 - Yes but not every feature, will use for few days but forget about it after a few days or lazy to upload
- Comments on wraps feature
 - For music part, separate into separate slides
 - Trends part, make photos into a collage
 - Rest looks good

3. Emily

- On a scale of 1 to 5, rate the general aesthetic
 - Like the colours and layout
 - The add entry button too invasive, should put at bottom
 - Too much empty space
 - Journal entry page look like android notes app
 - Like how related tools are grouped together
 - Score 4
- On a scale of 1 to 5, how easy was it to complete the tasks? Any tasks that were not intuitive?
 - Score 3
 - Viewing wraps not intuitive cos navigation icon is a tree
 - Thought the calendar colours is based on number of journal entries per day
- Would you personally use the app
 - Knows people who are very dedicated to spotify, feels that those people will appreciate it
 - Personally not that kind of person
 - Very interested in the wrap
- Anything you would like to see in the wraps? Fun facts? Graphics?
 - Like the genre, type, videos and summary pages
 - Okay with current wrap features

4. Maru

- On a scale of 1 to 5, rate the general aesthetic
 - Really liked the colour scheme, said the overall design was okay given that it was a lofi design.
 - Score 4

- On a scale of 1 to 5, how easy was it to complete the tasks? Any tasks that were not intuitive?
 - Score 3
 - Felt that it was difficult to do the wraps task as it was not obvious.
- Would you personally use the app
 - He said he would because he would like to find out more about his music taste
- Anything you would like to see in the wraps? Fun facts? Graphics?
 - He wanted a feature where you would match with similar users

5. Danny

- Struggled to upload photo, because it said upload file, so it was ambiguous
- Struggled to click the music icon because it did not look clickable
- Tried to click near notifications for settings
- Was not sure what to click for the wraps and the entries.
- On a scale of 1 to 5, rate the general aesthetic
 - Felt that the green color in the wraps was too jarring.
 - Felt that the color scheme was not very calming for a journaling app
- On a scale of 1 to 5, how easy was it to complete the tasks? Any tasks that were not intuitive?
 - He also felt that many design elements which were clickable were not obviously clickable.
 - Felt that settings should be near notifications because most apps are normally like that
- Would you personally use the app
 - He said he would if he did not need to add many things. He would be more interested in the photo aspect than the music aspect, though
- Anything you would like to see in the wraps? Fun facts? Graphics?
 - He thought that the color could be better. He liked the personality wraps thing. Otherwise he thought it was ok.

Testing Iteration 2

1. Jobelle

- Journal - Need to add a button to add song
- Wraps - didn't know it was wraps page
- Account - Didn't know can scroll to choose theme
- Calendar date selection confusing
- Calendar page prefer to see moods instead of normal
- Like to also include auto-predict mood
- Graphic design and aesthetic
 - Clean n like relaxing colours
 - Ability to customise

- Prefer light mode cos colours pop out more
- Would you personally use the app
 - Very lazy, wont use for long (3 days - 1 week)
- Any task hard to do/not intuitive
 - Calendar use popup instead
- Wraps
 - show comparisons w other people
 - Genres and music recommendations
 - Cats n dogs use percentage instead

2. Kenny

- Didn't know how to add music to journal
- Didn't know which one is wraps
- Journaling on select date scroll down
- Aesthetic
 - Icons are minimalistic enough
 - Want to add custom themes
- Complete task
 - Intuitive except add song to journal
- Didn't notice wraps share button, colours too nice
 - Share in last page
- Good to have wraps for the year
- Personally use app?
 - Use if more social aspect
- Recent memories: separate by date

3. Alphonsus

- Done journal entry properly
- Can view wraps with no problem
- Can change theme
- Can change view of calendar by mood/entry
 - Pop up more intuitive for calendar
 - Red more aggressive, lighter more calming
- Simplistic
- User friendliness?
 - Pretty great
- Difficult task to complete
 - View entries for the day
- Would you use it?
 - Yes, store everything in one space
- Wrap, what do you like?
 - Full screen, progress bar good

- Share button too big, can be in the first one
- Like stats for the music
- Nothing that they don't like
- Want to see photos tag by location, a heat map

4. Jenifer

- Done journal entry properly
- Viewed wraps, images seem a bit blur
- Has trouble navigating to view certain entry for the date
- Can change theme
- Change view (mood, no. of entries) of calendar, can do it
 - Popup or scroll down, both also can
- Rate 1-5 general aesthetic: 5
 - Colours quite calm, pastel, not affect mood soft to the eyes, not eye piercing
- Ease of use:
 - Quite intuitive except finding today's entry
- Anything you struggle with?
 - No
- Mood colours: no idea what they mean
- No. of entries: darker more entry
 - Would be better if its the same colour adjust brightness
- What did you like about the wraps:
 - Good idea, about giving persona
 - Weird that using mbti image is confusing (seems like our personality is changing every month)
 - More summary
 - Watch wraps section by section
- Would you personally use this kinds of app:
 - Would use but not sure how often
 - Likes monthly wraps
 - List of journal entry in calendar
 - filter by mood
 - Let user choose mood, as using AI makes it seem like there are other people looking at what user is writing
 - Wants the app to be personal
 - No journals in monthly wraps \

5. Danny

- Didn't know that the button looked like wraps
- Liked the overall aesthetic, rated it a 5
- Liked the fake mBti

- Okay with popup or dropdown, drop down is okay but pop up is more intuitive but harder to use,
- Can't go back on summary page for wraps if overclick.
- Would use if friends use, but not otherwise
-

6. Yong Jie

- Felt that it was intuitive that the pictures are at the bottom
- Felt that the aesthetic was cozy and liked clicking it
- Wouldn't know what the icons mean, especially, the wraps
- Felt that it was not difficult to use
- Felt that it made sense that the colors correspond to the moods in the front page
- Felt that it wasn't clear that darker entries represented the number entries, but felt that opacity would be more important.
- Would use this app only if his friends are using.
- Felt the Let's Go page in the summary was not necessary
- Would have no idea what to do with the information given in the wraps.

Appendix C: Metric Evaluation Tests

The following tasks were given for users to perform:

- Create a journal entry, adding a photo and song. (Minimum 6 clicks)

View Wraps. (Minimum 3 clicks)

View and edit an existing entry (Minimum 2 clicks)

View Settings (Minimum 1 click)

Update Theme (Minimum 2 clicks)

1 . Chang Zheng

Completed tasks on first try?

View Wraps. **No**

View and edit an existing entry **Yes**

View Settings **Yes**

Update Theme **Yes**

Satisfaction: 4.5/5

System Usability Scale

I think that I would like to use this system frequently. (3)

I did not find the system unnecessarily complex. (5)

I thought the system was easy to use. (5)

I think that I would not need the support of a technical person to be able to use this system. (5)

I found the various functions in this system were well integrated. (5)

I did not think there was too much inconsistency in this system. (5)

I would imagine that most people would learn to use this system very quickly. (5)

I did not find the system very cumbersome to use. (5)

I felt very confident using the system. (5)

I did not need to learn a lot of things before I could get going with this system. (4)

Usability Score 47/50 (More is better)

2. Brenda

Completed tasks on first try?

View Wraps. **Yes**

View and edit an existing entry **No**

View Settings **Yes**

Update Theme **Yes**

Satisfaction: 4/5

System Usability Scale

I think that I would like to use this system frequently. (1)

I did not find the system unnecessarily complex. (5)

I thought the system was easy to use. (5)

I think that I would not need the support of a technical person to be able to use this system. (5)

I found the various functions in this system were well integrated. (4)

I did not think there was too much inconsistency in this system. (5)

I would imagine that most people would learn to use this system very quickly. (5)

I did not find the system very cumbersome to use. (4)

I felt very confident using the system. (5)

I did not need to learn a lot of things before I could get going with this system. (5)

Usability Score 44/50

3. John

Completed tasks on first try?

View Wraps. **Yes**

View and edit an existing entry **Yes**

View Settings **Yes**

Update Theme**Yes**

Satisfaction: 5/5

System Usability Scale

I think that I would like to use this system frequently. (2)

I did not find the system unnecessarily complex. (5)

I thought the system was easy to use. (5)

I think that I would not need the support of a technical person to be able to use this system. (5)

I found the various functions in this system were well integrated. (4)

I did not think there was too much inconsistency in this system. (4)

I would imagine that most people would learn to use this system very quickly. (5)

I did not find the system very cumbersome to use. (4)

I felt very confident using the system. (5)

I did not need to learn a lot of things before I could get going with this system. (4)

Usability Score: 43/50

Appendix D: Usability Heuristics

10 Usability Heuristics



Visibility

Show system status, tell what's happening



Mapping

Use familiar metaphors & language



Freedom

Provide good defaults & undo



Consistency

Use same interface and language throughout



Error Prevention

Help users avoid making mistakes



Recognition

Make information easy to discover



Flexibility

Make advanced tasks fluid and efficient



Minimalism

Provide only necessary information in an elegant way



Error Recovery

Help users recognize, diagnose and recover from errors



Help

Use proactive and in-place hints to guide users

Based on Nielsen's ten heuristics. Updated by Scott Klemmer and Janaki Kumar.

Heuristics used from <https://www.nngroup.com/articles/ten-usability-heuristics/>

Appendix E: Style Guide

